

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11: 1.2 ECOLOGY – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: Why are fish disappearing from a water body that once supported many fish, and what can the community do to restore the ecosystem? In the Kenyan river video (2:20 to 4:50), a river that once had many fish now yields far fewer, while algae and aquatic weeds grow thickly and the water sometimes smells unpleasant. Use what you learned in Lessons 1 to 9, your notes, your presentations, your nitrogen and carbon cycle drawings and your pollution summary to answer each question below. Write in full sentences in the space under each prompt. Use correct ecology terms (for example ecosystem, habitat, population, density, biotic factor, abiotic factor, carrying capacity) and give evidence from our investigations. Where a question asks for a calculation, show your working.

Part 1: The river as an ecosystem and who thrives in it

1. In Lesson 1 some learners said the fish were disappearing because they are old or have no food. Use the ecological terms you learned (ecosystem, habitat, community, biotic factors, abiotic factors) to describe the river. Then explain why 'the fish are old' or 'they have no food' cannot alone explain what you see in the video.
2. Think about the adaptations you studied in Lesson 2 (for example water hyacinth and tilapia). Explain why the conditions in the river can favour algae and aquatic weeds while harming fish.

Write your answer here:

Part 2: Measuring populations

3. Define population and population density. In Lesson 3 you counted organisms in two 10 m x 10 m areas of the school compound. Explain why direct counting worked there but would be hard for the fish in the river.
4. In Lessons 4 and 5 you used quadrats and a belt transect. Describe what you found about how plants are spread in the school field and across the compound, and explain what caused the differences.
5. In Lesson 6 you marked 20 counters out of 100, then took a second sample of 20 and found 5 marked. Calculate the estimated population size and compare it with the true value. State two assumptions the method makes and explain how it could help the community find out whether fish numbers are falling.

Write your answer here:

Part 3: Factors that limit populations

6. Use your summary of the readings and the PhET simulation from Lesson 7. Explain what biotic and abiotic limiting factors are and what carrying capacity means. Give examples of each type of factor that could affect the river fish.

7. In the discussion diamond you ranked nine factors. Choose the factor you ranked highest and explain, using evidence, how well it explains the fish decline. Explain why one factor alone is probably not enough.

Write your answer here:

Part 4: How nutrients move and what pollution does

- 8. Draw and label a diagram of the nitrogen cycle and the carbon cycle, as you did in Lesson 8, and add a pathway showing how excess nutrients enter the river. Name at least three processes in your diagram.**
- 9. Using your pollution summary from Lesson 9, write a cause-and-effect explanation (a chain of steps) of how effluent, garbage and fertiliser runoff lead to fewer fish, thick algae and weeds, and a bad smell in the river.**
- 10. Name two types of pollution you recorded at the shopping centre. For each, say how it could end up in a river or affect the environment. Then explain which of the pollution types you recorded or researched contribute to global warming and climate change, and why.**

Write your answer here:

Part 5: Restoring the river

- 11. Compare your Lesson 1 model of the river with your final model. Describe two ideas that changed or were added.**
- 12. Write your Final Explanation: why are fish disappearing from the river that once supported many fish? Then propose at least three actions the community can take to restore the ecosystem, and say how each action links to a cause you identified.**

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of ecological concepts and terms	Correctly and consistently uses ecosystem, habitat, population, density, biotic and abiotic factors, carrying capacity and adaptation to explain the river.	Uses most terms correctly with minor errors or gaps.	Uses few terms, or uses them incorrectly.
Population measurement and calculations	Accurately explains direct counting, quadrat and transect sampling, and correctly calculates the capture-recapture estimate (80) with working, assumptions and comparison to 100.	Explains the methods and calculates correctly with minor omissions in working or assumptions.	Calculation is incorrect or methods are confused or not explained.
Causal explanation of the phenomenon	Gives a complete chain: nutrient and pollution input, algal and weed growth, decay by bacteria, oxygen loss, fish death or movement, smell, plus interacting limiting factors, using evidence from the lessons.	Gives most steps of the chain with some evidence.	Gives only isolated ideas or does not link steps.
Models and use of evidence	Nutrient cycle diagram is accurate and labelled with a clear pollution pathway; refers specifically to class investigations, the video and the pollution findings.	Diagram or evidence is mostly accurate and specific.	Diagram or evidence is missing, vague or inaccurate.
Community solutions	Proposes at least three realistic Kenyan community actions, each linked to a specific cause.	Proposes two or three actions, with some links to causes.	Proposes one or no actions, or they are not linked to causes.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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